

Assignment on NLP and Sentiment Analysis

Online Review Sentiment Analysis: Flipkart and Twitter

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GitHub repository: <https://github.com/Shoily01/-NLP-and-Sentiment-Analysis>

1. Introduction

Every day, thousands of short reviews of customer opinions are written about products on shopping sites and on social media. In this project, we teach a computer to read such a review and label it as positive, neutral or negative, and then examine how opinions evolve over time, between Flipkart and Twitter, and between different products. The data is generated synthetic of 1000 reviews from 15 products over a period of 12 months from Flipkart and Twitter (both data sources January to December 2025). The following workflow is used: review the data, preprocess the text and convert it to a numerical format (TF-IDF), partition the data, train and compare three classification models, analyze the model's errors, and then visualize the sentiment trends. The main lesson is as follows: The distribution of the data changes the results much more than the selection of the model, so it is important to do the testing honestly.

2. Models and results

2.1 Data and preprocessing

There are no missing values in the data set. Forty percent of the data is positive, 34% is negative, and 26% is neutral (left side of Figure 1). The remainder of the work was influenced by two checks. First, there are only 42 sentence templates in the 1,000 reviews (14 per class); the only difference is the product name, and each sentence template is always labeled the same. Second, the rating of the star is not correlated with the sentiment label (correlation -0.005 ; Figure 1, right), and therefore only the review text is used.

The cleaning steps are: Lower casing, stripping away links, @mentions and # signs, stripping punctuation, tokenising using NLTK, stripping stopwords and finally TF-IDF (first 3000 terms). Negation words (no, not) were not removed as removing them would make “not happy” into “happy” and the average F1 value would increase from 0.724 to 0.761, and single words instead of word pairs were used because there are too few templates for word pairs to be specific (0.761 vs 0.735).

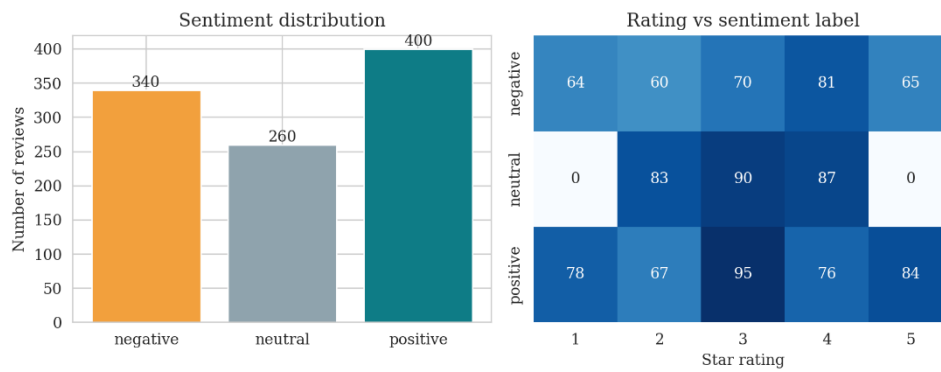


Figure 1. . Sentiment distribution (left) and star rating against sentiment label (right)

2.2 Models and comparison

Three models were trained: Logistic Regression, Multinomial Naive Bayes, both unregularised and tuned versions (tuned by grid search) of a Linear SVM, a “majority class” baseline model and a tuned version of each of these models (tuned by grid search). Two sets of reviews that had not been encountered in the sentence templates were tested with each: one hold-out set (196 reviews; 8 sentence templates) and repeated grouped cross-validation (3×5 folds). The best model was selected based on the F1 score that was calculated using cross-validated data, rather than the single hold-out score, which is noisy.

Table 1. Model comparison on unseen sentence templates (weighted scores)

Model	Accuracy	Precision	Recall	F1 (hold-out)	CV F1 (mean $\pm$ sd)
Linear SVM ✓ best	0.806	0.872	0.806	0.807	0.804 $\pm$ 0.090
Linear SVM (tuned)	0.801	0.870	0.801	0.802	0.800 $\pm$ 0.104
Logistic Regression (tuned)	0.699	0.741	0.699	0.706	0.770 $\pm$ 0.088
Logistic Regression	0.755	0.851	0.755	0.749	0.748 $\pm$ 0.058
Multinomial Naive Bayes	0.745	0.826	0.745	0.731	0.730 $\pm$ 0.059
Naive Bayes (tuned)	0.684	0.712	0.684	0.678	0.723 $\pm$ 0.102
Majority baseline	0.378	0.143	0.378	0.207	0.229 $\pm$ 0.013

The Linear SVM outperformed Logistic Regression (0.749) and Naive Bayes (0.731), and all performed well compared to the baseline (0.21). The ordering is also consistent for the SVM which ranked first in all four cleaning and n-gram settings. However, this is not the whole story as the gap is less than the fold-to-fold spread (approx. 0.09), which means that it's a probable advantage, but not a sure one. One possible explanation is that the features of TF-IDF are sparse, and are almost linearly separable, making SVM appropriate. However, tuning did not work well on a consistent basis: the tuned SVM and Naive Bayes classifiers performed no better than their defaults and the tuned Logistic Regression suffered on the hold-out but not in the CV. The linear SVM was thus retained as the default model. Its whole pipeline (cleaning, TF-IDF and model) was saved in one file, which was reloaded and which could sensibly label all 7

new test sentences, including the new sentences “Not good at all, it stopped working after two days” (negative).

2.3 Error analysis

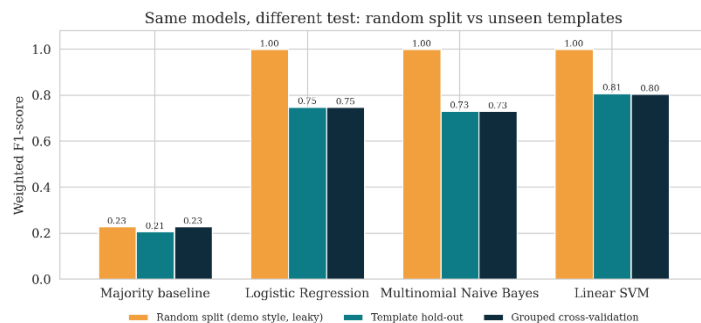


Figure 2. Same models, random split vs unseen templates

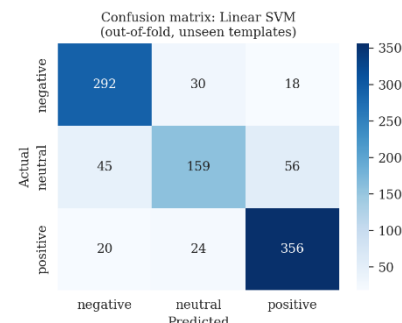


Figure 3. Confusion matrix, Linear SVM

Recall is good (0.89) for positive reviews, fair (0.86) for negative, and low (0.61) for neutral reviews (Figure 3), with 45 neutral reviews being either given a negative grade or 56 positive reviews being given a negative grade. This is natural since mild praise (works fine), mild complaints (packaging could be improved), and sentences that contain no opinion word (Comparing a few options before I decide) are all combined. There are also errors that form blocks - in 8 of the 42 templates, the model was incorrect in all reviews, because the wording was not at all similar to the other templates. That is, the model learned a small dictionary of useful words rather than a general sentiment. The language used should make sense, such as “not”, “broken” and “unacceptable” for negative and “amazing”, “exceeded” and “best” for positive.

3. Why a random split is misleading

In each of the class demos, the random splits were 80/20, with every model getting a perfect F1 score of 1.00 (orange bars in Figure 2). This is because each template occurred approximately 24 times in the data: 76% of the test reviews had an exact match in the training data, and all of the test reviews had a template that matched at least one of the training reviews. This test only measures memory, and it doesn't know if the model is good or bad. When a grouped split is held out (whole templates, as is seen in real new reviews), the score drops to 0.73–0.81, with a noticeable difference between the models. The same principle applies to splitting by time for forecasting: the test data should be similar to the data that will be encountered in the future, not the training data.

4. Sentiment trends

Positive reviews are 30–47% of all reviews each month, and are fairly consistent ($p = 0.60$) across 2025, with a maximum of 36 reviews in March. When it comes to Flipkart and Twitter, they are almost similar in the aspect of positive sentiment (40% on both) and negative sentiment (33% against 35%, $p = 0.8$). Tweets are different words used (mentions, hashtags, questions) but not negative or positive in non-discriminatory ways. By product (Figure 5, right), the most negative (net -13 points) are Air Fryer while the most positive (+29 and +26 points) are Running Shoes and Power Bank, this is not statistically significant ($p = 0.40$) with approximately 66 reviews per product. The data and dates are made-up and should not be interpreted as actual market analysis.

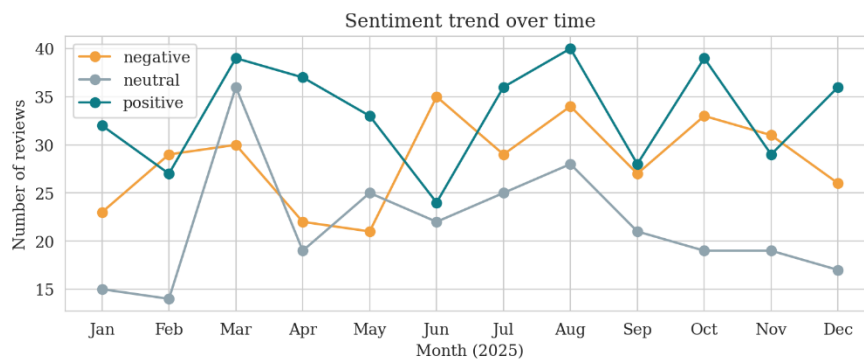


Figure 4. Sentiment trend over time

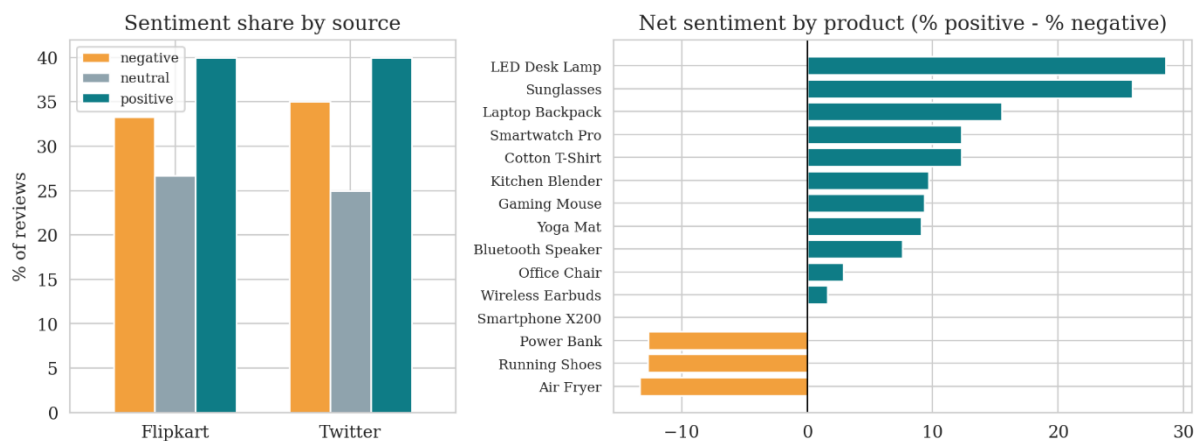


Figure 5. Sentiment by source and by product

5. Limitations and improvements

- Synthetic, template-based data. There are only 42 sentence patterns so they will not be carried across to real reviews. Improvement: conduct a study on authentic Flipkart and Twitter page reviews.
- Small, noisy hold-out. The tight test set has only 8 templates and single scores are unstable (fold spread ~ 0.09). Improvement: Report the main result, cross validation, as in this case and include additional data.

- Unreliable rating and lack of realistic language. No sarcasm, no emoji's, no spelling errors and no mixed languages, ratings don't match the labels. Improvement: see how the labels were fabricated and add some text.
- Bag-of-words models. TF-IDF does not consider order or sense of the words. Improvement: fine-tune a pre-trained transformer (e.g. BERT), that works much better with new wording, and also with neutral text.

6. Conclusion

The use of a Linear SVM on TF-IDF features yielded the best performance (F1 near 0.80 on unseen wording) and retaining negation words was the most helpful preprocessing option. The ideal splits of a random split were an illusion due to repetition of sentence templates, which is more important than the split with the model. In this synthetic data, sentiment did not meaningfully vary across month, source or product and there is no particular sentiment in neutral reviews.

The End